Algebra 1 Name____________________
Unit 11 Date____________________
Lesson 4 Homework Period__________________

Use the scenario below for questions 1 – 5.

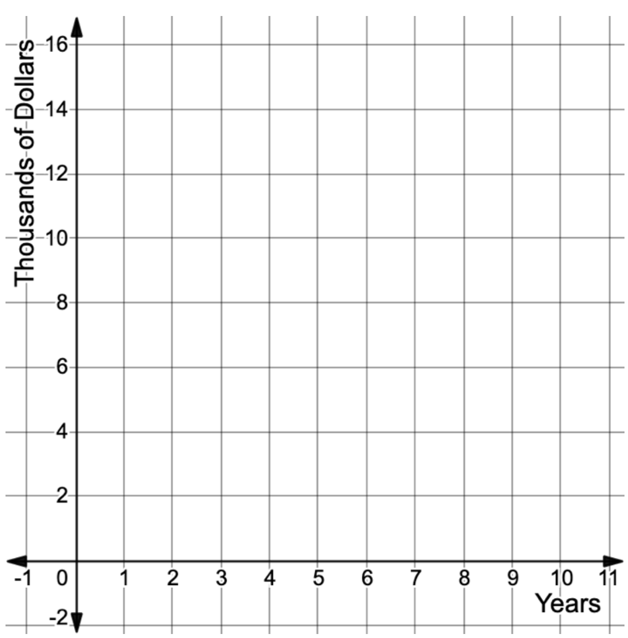
Tom purchased a used Honda CRV from a dealership for $\$15{,}575$. The car depreciates 15% per year. The situation can be modeled by the function $P(x) = \$15{,}575(0.85)^x$ where x represents the number of years since Tom purchased the car and P is the value of the car.

1. Does the equation $P(x) = \$15{,}575(0.85)^x$ represent exponential growth or decay?

2. What is the y-intercept and what does it represent in this situation?

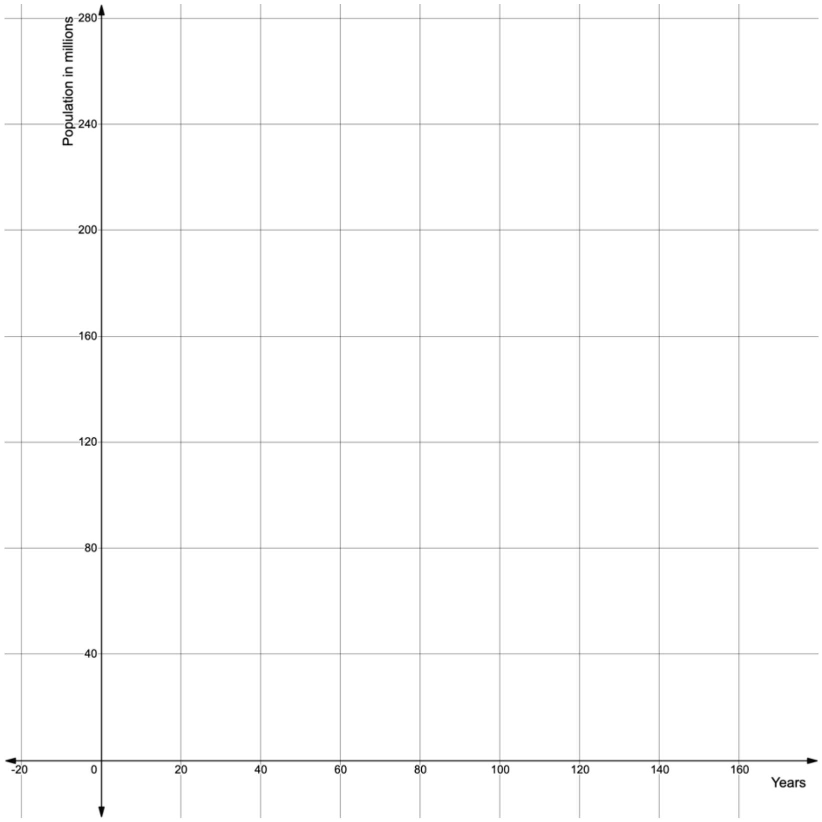
3. Explain why the value of b is not 0.15, the rate of depreciation.

4. Graph the equation $P(x) = \$15{,}575(0.85)^x$.



The equation $f(t) = 76(1 + 0.013)^t$ represents the U.S. population and t represents the number of years after 1900. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



Key Features		Interpretations
6. Domain		
7. Range		
8. Y-intercept		
9. Constant Percent Rate of Change		
10. End Behavior		